ALGEBRAIC RESIDUE FIELD EXTENSIONS LAND IN CLOSED POINTS OF A DECENT ALGEBRAIC SPACE

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 3587-3600, immediately following the statement at 3572-3585.

Editorial note. One cross-chapter pointer (Morphisms) is recorded in assembly.external_references; the decency input is a same-chapter line-indexed stub.

Editorial note. Related scheme-side statement morphisms-lemma-algebraic-residue-field-extension-closed-point-fibre is the imported criterion, not a competing count (different chapter and hypothesis).

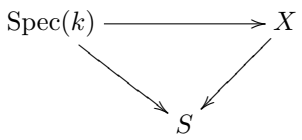
Allowed standard prerequisites (editorial).

- Specialization lifting in decent spaces. Lemma lemma-decent-specialization of this chapter, cited at proof lines 3591-3592. Lets any specialization x ~ x' in |X| be lifted to a specialization u ~ u' in an etale cover U with u over x.
- Closed points detected by algebraic residue field extensions on schemes. Morphisms, Lemma morphisms-lemma-algebraic-residue-field-extension-closed-point-fibre, cited at proof lines 3597-3599. The scheme-side criterion: a point whose residue field is algebraic over the residue field of a closed point below is closed.

Source reference. Original lemma 03IL: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 2878–2878. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01TE: focused-seeds/geometry/S9-decent-algebraic-residue-field-closed-point/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let X be a decent algebraic space over S. Consider a commutative diagram*



Date: stacks-project:decent-spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Assume that the image point $s \in S$ of $\mathrm{Spec}(k) \to S$ is a closed point and that $\kappa(s) \subset k$ is algebraic. Then the image x of $\mathrm{Spec}(k) \to X$ is a closed point of $|X|$.

Proof. Suppose that $x \rightsquigarrow x'$ for some $x' \in |X|$. Choose an étale morphism $U \to X$ where U is a scheme and a point $u' \in U'$ mapping to x'. Choose a specialization $u \rightsquigarrow u'$ in U with u mapping to x in X, see Lemma 03IL. Then u is the image of a point w of the scheme $W = \mathrm{Spec}(k) \times_X U$. Since the projection $W \to \mathrm{Spec}(k)$ is étale we see that $\kappa(w) \supset k$ is finite. Hence $\kappa(w) \supset \kappa(s)$ is algebraic. Hence $\kappa(u) \supset \kappa(s)$ is algebraic. Thus u is a closed point of U by Morphisms, Lemma 01TE. Thus $u = u'$, whence $x = x'$. □